DIABETES RESEARCH AND CLINICAL PRACTICE XXX (2014) XXX–XXX

Contents available at ScienceDirect

Diabetes Research and Clinical Practice

journal homepage: www.elsevier.com/locate/diabres

Role of various indices derived from an oral glucose tolerance test in the prediction of conversion from prediabetes to type 2 diabetes

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ARTICLE INFO

Article history:
Received 22 May 2014
Received in revised form
16 July 2014
Accepted 23 August 2014
Available online xxx

Keywords:
β-Cell function
Insulin resistance
Oral glucose tolerance test
Prediabetes
Type 2 diabetes

ABSTRACT

Aims: The clinical implications of prediabetes for development of type 2 diabetes may differ for Asian ethnicity. We investigated various indices derived from a 2-h oral glucose tolerance test (OGTT) in people with prediabetes to predict their future risk of diabetes.
Methods: We recruited 406 consecutive subjects with prediabetes from 2005 to 2006 and followed them up every 3–6 months for up to 9 years. Prediabetes was defined as isolated impaired fasting glucose (IFG), isolated impaired glucose tolerance (IGT), combined glucose intolerance (CGI), or isolated elevated HbA1c (5.7–6.4%, 39–46 mmol/mol) without IFG or IGT. The rate of diabetes conversion was compared between prediabetes categories. The association of glycemic indices with development of diabetes was also investigated.
Results: Eighty-one patients were diagnosed with diabetes during the 9-year follow-up (median 46.0 months). The rate of diabetes conversion was higher in subjects with CGI (31.9%), or isolated IGT (18.5%) than in those with isolated IFG (15.2%) or isolated elevated HbA1c (10.9%). Surrogate markers reflecting β-cell dysfunction were more closely associated with diabetes conversion than insulin resistance indices. Subjects with a 30-min postload glucose ≥165 mg/dL and a 30-min C-peptide <5 ng/mL had 8.83 times greater risk (95% confidence interval 2.98–26.16) of developing diabetes than other prediabetic subjects.
Conclusions: In Asians, at least Koreans, β-cell dysfunction seems to be the major determinant for diabetes conversion. A combination of high glucose and low C-peptide levels at 30 min after OGTT may be a good predictor for diabetes conversion in this population.
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http://dx.doi.org/10.1016/j.diabres.2014.08.014
0168-8227/© 2014 Published by Elsevier Ireland Ltd.

1. Introduction

The prevalence of prediabetes is increasing, and 35% of adults aged 20 years or older and 50% of people aged 65 years or older in the United States are estimated to have prediabetes [1]. This will lead to a future increase in type 2 diabetes. Prediabetes status is defined as intermediate hyperglycemia that shares the main pathophysiology of diabetes: impaired insulin secretion and increased insulin resistance [2]. The definition of prediabetes has been revised over time [3]. In the 2010 American Diabetes Association (ADA) guidelines, a hemoglobin A_{1C} (HbA1c) of 5.7–6.4% (39–46 mmol/mol) has been included in a category of markers of increased risk of diabetes that also includes impaired fasting glucose (IFG) and impaired glucose tolerance (IGT) [4]. Several epidemiologic studies have shown that IFG and IGT describe distinct populations of hyperglycemia [5,6]. IFG reflects hepatic insulin resistance [7,8], while IGT reflects relatively high muscle insulin resistance [9]. Both IFG and IGT share impaired early-phase insulin secretion, while IGT also shows a reduction in late-phase insulin secretion [10]. Thus, individuals with prediabetes defined by distinct diagnostic criteria might be a heterogeneous group with different pathophysiology and prognosis.

In this situation, a more detailed assessment for predicting development of diabetes in people with prediabetes is needed. Older age and a family history of diabetes are well-known risk factors for this, but biochemical measures, such as lipids, uric acid, or liver enzymes, have failed to demonstrate predictive value for diabetes [11–13]. Fasting insulin concentration was also not notably useful for the prediction of diabetes [14]. However, measures derived from the oral glucose tolerance test (OGTT), including 1-h glucose or the insulin secretion/insulin resistance index showed strong predictive power for diabetes [15]. A recent study has shown that the patterns of insulin concentration during an OGTT strongly predict the development of diabetes [16]. Thus, the OGTT could provide more accurate information about the risk of diabetes, especially in people with prediabetes who have impaired glucose metabolism [16,17]. More specifically, plasma insulin and C-peptide concentrations after glucose loading are more likely to reflect β-cell dysfunction and insulin resistance than are fasting samples. Further, the combination of insulin and glucose levels at a specific time point during the OGTT might be a good predictor of future diabetes. However, to the best of our knowledge, no previous studies have assessed the predictive value of this combination.

In this longitudinal study, we followed subjects with prediabetes for up to 9 years for the development of diabetes. We compared the ability to predict conversion to diabetes between different definitions of prediabetes, and assessed various parameters related to glycemic regulation derived from the glucose and insulin concentrations detected in a three-point OGTT (0, 30, and 120 min). Finally, we suggest a diabetes risk-prediction model using a combination of glycemic and insulin-related parameters.

2. Subjects, materials and methods

The study population included pre-screened individuals with prediabetes who visited the diabetes clinic at Seoul National University Bundang Hospital (SNUBH), a tertiary referral hospital, in 2005 and 2006 for evaluation of their glycemic control. They were referred to the diabetes clinic after they were diagnosed with prediabetes at their health check-up or primary clinic. Subjects who took oral hypoglycemic agents or insulin were excluded from this study. We consecutively enrolled 418 participants with prediabetes based on the results of a 75-g OGTT. After excluding 12 who were lost to follow-up within 1 year after enrollment, 406 participants were included with a follow-up rate of 97.1%. The median follow-up duration was 46.0 (3.0–107.8) months. Among 406 participants with prediabetes, 141 subjects agreed to measure all the values of plasma C-peptide, insulin, and glucose at fasting, 30 min, and 2 h. Predictive model for type 2 diabetes using 30-min plasma glucose and 30-min C-peptide was based on this data. This study was approved by the institutional review board of SNUBH. Written informed consent was obtained from every participant.

2.1. *Definition of prediabetes and diabetes*

Prediabetes was defined according to the ADA 2010 criteria [4]. Isolated IFG was defined as fasting plasma glucose (FPG) levels of 100–125 mg/dL and a 2-h postprandial glucose (2-h PG) <140 mg/dL, and isolated IGT as 2-h PG 140–199 mg/dL and FPG <100 mg/dL. Subjects with IFG and IGT were defined as combined glucose intolerance (CGI). Subjects with HbA1c of 5.7–6.4% (39–46 mmol/mol) and neither IFG nor IGT were also included in this study as a category of isolated elevated HbA1c prediabetes. Diabetes mellitus was diagnosed when the FPG was ≥126 mg/dL, or the 2-h PG was ≥200 mg/dL or HbA1c was ≥6.5% (48 mmol/mol).

2.2. *Measurement of anthropometric and biochemical parameters*

Height and body weight were measured by standard methods. BMI was calculated as body weight divided by height squared (kg/m^2). Blood pressure measurements were made after subjects had remained seated for 10 min. Measurements were made twice, with a 5-min rest period in between, and the mean value of the measurements was used. Smoking status was divided into three categories: current smokers, ex-smokers, and never smokers. Alcohol intake was assessed by frequency and quantity of beer, spirit, sake, and wine intake during the previous 12 months, and was categorized into non-, light (<1/wk), moderate (≤3/wk), and heavy (>3/wk) drinkers. Physical activity was classified into four categories: no, light (1/wk), moderate (2–3/wk), and heavy (≥4/wk) exercise episodes. One episode of exercise was defined as exercising for at least 30 min.

Plasma glucose levels were measured with a Hitachi 747 chemistry analyzer (Hitachi, Tokyo, Japan). HbA1c was measured with a Bio-Rad variant II Turbo HPLC analyzer (Bio-Rad, Hercules, CA, USA) in the National Glycohemoglobin Standardization Program (NGSP) level II certified laboratory at SNUBH. Plasma C-peptide and insulin concentrations were measured by radioimmunoassay (Linco, St. Louis, MO, USA). The fasting plasma concentration of total cholesterol, triglycerides, and high-density lipoprotein (HDL) and low-density

lipoprotein (LDL) cholesterol were measured using the Hitachi 747 chemistry analyzer.

2.3. Indices of β-cell function or insulin sensitivity

The homeostasis model assessment (HOMA) was used to measure insulin resistance and β-cell function: HOMA-IR and HOMA-β [18]. The formula for HOMA-IR is as follows: fasting insulin (μU/mL) × [fasting glucose (mg/dL)/405]. HOMA-β was calculated using the following formula: 360 × fasting insulin (μU/mL)/[fasting glucose (mg/dL) − 63]. The Stumvoll index was assessed for first- and second-phase insulin with the following equations: first-phase insulin release = 1283 + 1.829 × 30-min insulin (μU/mL) − 138.7 × 30-min glucose (mmol/L) + 3.772 × fasting insulin (μU/mL); second-phase insulin release = 287 + 0.4164 × 30-min insulin (μU/mL) − 26.07 × 30-min glucose (mmol/L) + 0.9226 × fasting insulin (μU/mL) [19]. The Matsuda index was calculated as 10,000/square root of [fasting glucose (mg/dL) × fasting insulin (μU/mL)] × [mean glucose (mg/dL) × mean insulin (μU/mL) during an OGTT] [20]. Mean glucose and insulin levels were calculated from values at 0, 30, and 120 min of the OGTT, using the trapezoidal rule [20]. The insulinogenic index (30′), as an indicator of early-phase insulin release, was calculated by [30-min insulin − fasting insulin]/[30-min glucose − fasting glucose] [21]. The insulinogenic index (120′) was calculated using mean insulin/mean glucose or [area under the curve of insulin]/[area under the curve of glucose] using trapezoidal integration at 0, 30, and 120 min. The disposition index, which is known to reflect β-cell function, was calculated as [30-min insulin − fasting insulin]/[30-min glucose − fasting glucose]/HOMA-IR [22,23]. The fasting glucose to insulin ratio was also measured to assess insulin sensitivity [24]. The quantitative insulin sensitivity check index (*QUICKI*) was calculated as 1/[log (fasting insulin (μU/mL)) + log (fasting glucose (mg/dL))] [25].

2.4. Statistical analysis

Variables are presented as means ± SD or *n* (percent). Baseline characteristics of subjects were described for the following categories of prediabetes: isolated IFG, isolated IGT, CGI, or isolated elevated HbA1c prediabetes. ANOVA was used to compare the mean differences. The Scheffe and Duncan methods were used to perform *post hoc* analysis. The χ^2 test was used to compare the differences between categorical variables. Kaplan–Meier survival analysis was carried out to compare the median duration before the development of diabetes for the participants with isolated IFG, isolated IGT, CGI, and isolated elevated HbA1c prediabetes. The durations of conversion to diabetes for each category of prediabetes were compared using the log-rank test. We also analyzed various parameters obtained during the OGTT that showed significant differences between converters to diabetes and nonconverters. Correlation analysis was performed to select independent OGTT-related indices for the prediction of incident diabetes. Glycemic parameters FPG, 30-min PG, 2-h PG, and HbA1c were further analyzed for their power to predict diabetes using receiver operating characteristic (ROC) curves. The diagnostic properties of the cutoff values of 30-min PG were evaluated with the Youden index, defined as [sensitivity + specificity − 1] [26]. C-peptide at 30 min and insulin-secretion-related indices showed low AUC values; hence, we used multiple logistic regression analysis to define the cutoff point of diabetic conversion, adjusting for age, sex, BMI, family history of diabetes, and smoking. For each category based on the 30-min PG and 30-min C-peptide, the relative risk for incidence of diabetes was calculated using Cox proportional hazard analysis after adjustment for age, sex, BMI, family history of diabetes, and smoking. β-Cell function and insulin sensitivity-related indices in these groups were compared using analysis of variance (ANOVA) or Kruscal–Wallis tests, if they were not normally distributed.

3. Results

3.1. Baseline characteristics of the study subjects

Table 1 presents the baseline characteristics of the study population. Of 406 participants, 158 were defined with isolated IFG, 65 had isolated IGT, and 119 had both higher FPG and 2-h PG (CGI). Additionally, 64 subjects with abnormal HbA1c values despite normal FPG and 2-h PG were included as isolated elevated HbA1c prediabetes. The average participant age was 60.2 ± 10.8 years and the mean BMI was 24.6 ± 3.5 kg/m^2. HOMA-β was lower in the isolated IFG and CGI groups (81.6 ± 51.1 and 91.6 ± 43.8, respectively) than in those with isolated IGT or isolated elevated HbA1c (172.4 ± 149.8 and 170.6 ± 141.8, respectively). The CGI group showed higher HOMA-IR than the isolated elevated HbA1c group (3.1 ± 1.5 and 2.4 ± 1.2, $P = 0.004$). Other parameters, except HDL cholesterol and BMI, were similar between prediabetes groups.

3.2. Diagnostic criteria for prediabetes and the incidence of type 2 diabetes

During the median 46-month follow-up period (3.0–107.8 months), 81 participants were diagnosed with diabetes with a conversion rate of 20.0%. The cumulative incidence of diabetes was 10.9%, 15.2%, 18.5%, and 31.9% for isolated elevated HbA1c, isolated IFG, isolated IGT, and CGI, respectively ($P = 0.001$ by χ^2 test). The mean time to diabetes conversion was 44.1, 49.4, 40.0, and 35.2 months for isolated IFG, isolated elevated HbA1c, isolated IGT, and CGI, respectively ($P < 0.001$ by log-rank test, Fig. 1). The Kaplan–Meier curve shows a proportionally higher risk of diabetes progression in the CGI and the isolated IGT groups than in the isolated IFG or isolated elevated HbA1c groups (Fig. 1).

3.3. Glycemic parameters and indices derived from an OGTT in subjects who converted to type 2 diabetes

The glycemic and insulin-related parameters during the 2-h OGTT are compared between converters ($n = 81$) and nonconverters ($n = 325$) in Table 2. There was no difference in age, sex, family history of diabetes, BMI, smoking status, blood pressure, or lipid levels between the patients who progressed to diabetes and those who did not. Glycemic parameters including HbA1c and plasma glucose concentrations at 0, 30, and 120 min during the OGTT were higher in

Table 1 – Baseline characteristics of the study subjects.

	Total (n = 406)	Isolated IFG (n = 158)	Isolated IGT (n = 65)	Combined glucose intolerance (n = 119)	Isolated elevated HbA1c (n = 64)	P value
Age (years)	60.2 ± 10.8	60.2 ± 11.3	63.0 ± 11.0	59.1 ± 10.1	59.3 ± 10.1	0.12
Male sex (%)	49.3	51.3	43.1	52.1	43.8	0.49
Family history of type 2 diabetes (%)	22.4	29.1	13.8	21.8	15.6	0.04
BMI (kg/m^2)	24.6 ± 3.5	24.7 ± 3.0	23.2 ± 3.5*	25.1 ± 3.3‡	24.9 ± 4.7	<0.01
Waist circumference (cm)	87.6 ± 8.8	87.0 ± 8.9	85.7 ± 10.2	87.8 ± 7.9	90.3 ± 9.3	0.26
Smoking (%)						0.18
Current smoker	12.6	10.1	15.4	14.3	12.5	
Ex-smoker	14.8	19.0	7.7	16.8	7.8	
Alcohol intake (%)						0.04
Non drinker	69.7	70.9	73.8	59.7	81.3	
Light (<1/wk)	11.3	10.8	12.3	13.4	7.8	
Moderate (≤3/wk)	9.1	8.2	4.6	16.8	1.6	
Heavy (>3/wk)	9.9	10.1	9.2	10.1	9.4	
Physical activity (%)						0.04
No	57.6	52.5	75.4	52.9	60.9	
Light (1/wk)	9.6	12.7	1.5	13.4	3.1	
Moderate (2–3/wk)	14.5	17.1	10.8	11.8	17.2	
Heavy (≥4/wk)	18.2	17.7	12.3	21.0	18.8	
Blood pressure (mmHg)						
Systolic	128.3 ± 14.4	128.9 ± 14.2	126.6 ± 14.4	127.6 ± 14.8	128.2 ± 14.1	0.81
Diastolic	77.9 ± 10.9	78.0 ± 11.4	77.0 ± 10.4	77.9 ± 11.1	77.9 ± 9.6	0.97
Total cholesterol (mg/dL)	202.3 ± 40.0	206.3 ± 38.5	196.7 ± 46.9	198.4 ± 39.8	205.4 ± 36.7	0.23
Triglycerides (mg/dL)	145.5 ± 89.1	145.7 ± 93.6	150.1 ± 76.3	147.7 ± 80.3	136.5 ± 105.2	0.83
HDL-cholesterol (mg/dL)	52.9 ± 14.8	53.9 ± 13.3	47.6 ± 18.8*	51.6 ± 12.8‡	58.0 ± 15.6§	<0.01
LDL-cholesterol (mg/dL)	111.3 ± 30.6	112.5 ± 29.9	110.3 ± 39.3	110.3 ± 28.2	111.9 ± 27.0	0.92
AST (IU/L)[a]	25.3 ± 10.3	25.6 ± 10.4	22.9 ± 6.7	26.3 ± 8.2	24.5 ± 14.4	0.60
ALT (IU/L)[a]	29.1 ± 20.8	28.2 ± 20.9	23.8 ± 15.3	33.5 ± 23.5	27.4 ± 17.3	0.22
Creatinine (mg/dL)	1.0 ± 0.2	1.0 ± 0.2	0.9 ± 0.2	1.0 ± 0.2	1.0 ± 0.2	0.57
HOMA-β	113.9 ± 100.1	81.6 ± 51.1	172.4 ± 149.8†	91.6 ± 43.8‡	170.6 ± 141.8†	<0.01
HOMA-IR	2.8 ± 1.4	2.7 ± 1.6	2.5 ± 1.0	3.1 ± 1.5	2.4 ± 1.2§	<0.01

* $P < 0.05$ vs. Isolated IFG.
† $P < 0.001$ vs. Isolated IFG.
‡ $P < 0.01$ vs. Isolated IGT.
§ $P < 0.05$ vs. Combined glucose intolerance.
[a] AST and ALT, aspartate and alanine aminotransferase.

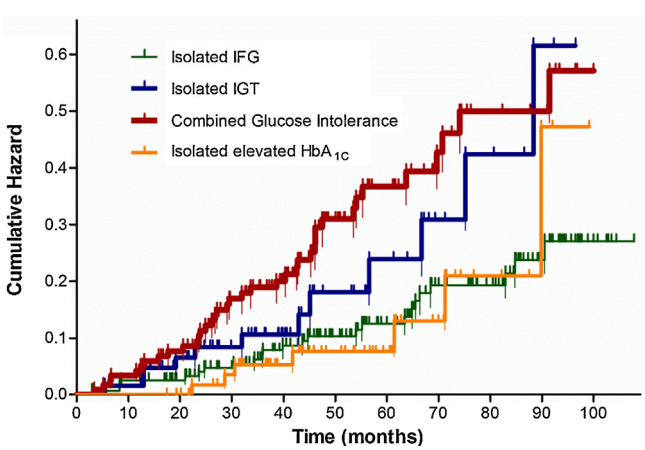

Fig. 1 – Kaplan–Meier survival curves showing the cumulative incidence of type 2 diabetes in the different prediabetes categories during the follow-up period ($P < 0.01$ by log-rank test).

converters than in nonconverters ($P < 0.001$). Fasting insulin and C-peptide levels did not differ between converters and nonconverters (Table 2). Of note, the insulin and C-peptide concentrations at 30 min showed significant differences between converters and nonconverters ($P = 0.027$ and 0.015, respectively).

Insulin sensitivity and indices of β-cell function are compared between converters and nonconverters in Table 2. The β-cell function-related indices were significantly lower in converters than nonconverters ($P < 0.01$ for all indices). There was no significant difference in insulin sensitivity-related indices. The disposition index, which reflects both β-cell function and insulin sensitivity, was lower in converters than in nonconverters ($P < 0.01$).

3.4. *Predictive model for type 2 diabetes using 30-min plasma glucose and 30-min C-peptide levels*

Among glycemic parameters, 30-min PG showed the strongest predictive power for diabetes with the highest AUC value (0.673, Table 3). The 165 mg/dL cutoff was selected as the best value for 30-min PG with the highest Youden index. Sensitivity

Table 2 – Glycemic parameters and insulin secretion-related indices derived from the OGTT and cardiovascular risk profile.

	Diabetes converters (n = 81)	Nonconverters (n = 325)	P value
Age (years)	62.3 ± 10.2	59.7 ± 10.9	0.06
Male sex (%)	42.0	50.8	0.17
Family history of type 2 diabetes (%)	21.0	22.8	0.88
BMI (kg/m^2)	24.8 ± 2.9	24.6 ± 3.7	0.40
Waist circumference (cm)	88.1 ± 9.0	87.4 ± 8.8	0.63
Smoking (%)			
current smoker	11.1	12.9	0.87
ex-smoker	16.0	14.5	
Blood pressure (mmHg)			
Systolic	129.5 ± 14.8	127.8 ± 14.3	0.40
Diastolic	78.8 ± 10.4	77.6 ± 11.0	0.44
Total cholesterol (mg/dL)	195.1 ± 34.9	204.1 ± 41.2	0.07
Triglycerides (mg/dL)	145.9 ± 90.8	144.3 ± 82.7	0.89
HDL-cholesterol (mg/dL)	53.9 ± 15.7	52.6 ± 14.6	0.49
LDL-cholesterol (mg/dL)	108.3 ± 26.1	112.1 ± 31.6	0.32
HbA1c (%)	5.97 ± 0.29	5.77 ± 0.33	<0.01
(mmol/mol)	(41.7 ± 3.2)	(39.5 ± 3.6)	
Plasma glucose (mg/dL)			
Fasting	107.2 ± 10.1	102.4 ± 11.5	0.01
30-min	189.1 ± 27.5	172.5 ± 27.2	<0.01
2-h	148.5 ± 32.7	132.9 ± 29.5	<0.01
Plasma insulin (μIU/mL)			
Fasting	11.08 ± 5.75	10.87 ± 5.52	0.77
30-min	37.34 ± 24.89	50.97 ± 30.72	0.03
2-h	57.84 ± 32.51	61.68 ± 53.22	0.61
Plasma C-peptide (ng/mL)			
Fasting	1.97 ± 0.87	1.89 ± 0.93	0.52
30-min	4.02 ± 2.18	5.37 ± 2.29	0.02
2-h	7.74 ± 2.99	8.96 ± 5.20	0.31
Stumvoll index			
First phase insulin release	631.8 ± 355.6	937.4 ± 525.4	<0.01
Second phase insulin release	194.5 ± 85.0	266.2 ± 119.6	<0.01
HOMA-β	95.1 ± 53.4	118.7 ± 108.4	<0.01
Insulinogenic index (30′)	0.3 ± 0.2	0.6 ± 0.4	<0.01
Insulinogenic index (120′)	0.25 ± 0.12	0.34 ± 0.21	<0.01
Disposition index	0.12 ± 0.08	0.21 ± 0.19	<0.01
Glucose to insulin ratio	12.2 ± 7.3	11.5 ± 5.7	0.40
HOMA-IR	2.9 ± 1.6	2.7 ± 1.4	0.24
QUICKI	0.33 ± 0.02	0.34 ± 0.02	0.31
Matsuda index[a]	4.6 ± 3.5	4.3 ± 2.4	0.58
Prevalence of coronary heart disease (%)	3.7	3.4	1.00

[a] Mean glucose, insulin and C-peptide were calculated from values at 0, 30, and 120 min of the OGTT, using the trapezoidal rule.

and specificity for 30-min PG ≥165 mg/dL were 91.8% and 38.6%, respectively.

Of the insulin and C-peptide concentrations derived from the OGTT, C-peptide at 30 min was selected because it has a lower P value than 30-min insulin (Table 2). Subjects with 30-min C-peptide <5 ng/mL showed the highest hazard ratio (8.36, 95% confidence interval [CI] 2.20–31.69, P = 0.002) for incident diabetes among several thresholds (Table 4). Hence, a 30-min C-peptide of 5 ng/mL was selected for the cutoff point in the diabetes risk-prediction model.

Table 3 – Glycemic parameters from OGTT in the prediction of diabetes among subjects with prediabetes.

Parameters	[a]AUC (95% confidence interval)	P value
Fasting plasma glucose	0.617 (0.540–0.695)	<0.01
30-min plasma glucose	0.673 (0.603–0.744)	<0.01
2-h plasma glucose	0.619 (0.535–0.704)	<0.01
HbA1c	0.647 (0.569–0.726)	<0.01

[a] AUC, area under the curve.

Based on the 30-min PG and 30-min C-peptide, 141 subjects were subcategorized into four groups (Fig. 2). Patients with 30-min PG ≥165 mg/dL and 30-min C-peptide <5 ng/mL (n = 39) showed 8.83 times greater risk of diabetes progression (95% CI 2.98–26.16) after multiple adjustment. Among the participants with 30-min PG ≥165 mg/dL and 30-min C-peptide ≥5 ng/mL who had significantly increased insulin resistance and modest β-cell function, 7.3% progressed to diabetes within 2–4 years (Table 5). The conversion rate to diabetes was 9.7% in the participants with 30-min PG <165 mg/dL and 30-min C-peptide <5 ng/mL, who had lower insulin resistance and decreased β-cell function. Of note, there was no converter in the group with 30-min PG <165 mg/dL and 30-min C-peptide ≥5 ng/mL.

Please cite this article in press as: Kim YA, et al. Role of various indices derived from an oral glucose tolerance test in the prediction of conversion from prediabetes to type 2 diabetes. Diabetes Res Clin Pract (2014), http://dx.doi.org/10.1016/j.diabres.2014.08.014

Table 4 – Multiple logistic regression analysis for the development of type 2 diabetes among subjects with prediabetes according to the 30-min C-peptide levels (*n* = 141).

Cutoff values of 30-min C-peptide (ng/mL)	*n*	*n*[a]	Hazard ratio[b]	95% confidence interval	*P* value
<2	7	2	2.76	0.43–17.79	0.29
<3	25	7	3.32	1.05–10.45	0.04
<4	46	12	5.23	1.73–15.87	<0.01
<5	70	17	8.36	2.20–31.69	<0.01
<6	95	17	3.61	0.97–13.46	0.06
<7	107	18	3.53	0.75–16.67	0.11

[a] Numbers of subjects who developed type 2 diabetes.
[b] After adjustment for age, sex, body mass index, family history of type 2 diabetes, and smoking.

4. Discussion

In this longitudinal study, subjects with isolated IGT or CGI had a higher risk of and shorter duration to diabetes conversion than those with isolated IFG or isolated elevated HbA1c. The subjects with 30-min PG ≥165 mg/dL and 30-min C-peptide <5 ng/mL had an 8.83 times greater risk of diabetes. In contrast, there was no converter in the group with 30-min PG <165 mg/dL and 30-min C-peptide ≥5 ng/mL.

The findings of the current study suggest that impaired β-cell function is the main pathophysiology in the development of diabetes in this prediabetes population. The β-cell-related indices were significantly decreased in diabetes converters, while insulin resistance-related parameters did not differ between converters and nonconverters. Several studies have shown that β-cell dysfunction is the major driving factor for the development of overt diabetes in Asians, Pima Indians, and Caucasians [27–32]. Differences in underlying β-cell function in each prediabetes category might contribute to the heterogeneous rate of progression to diabetes.

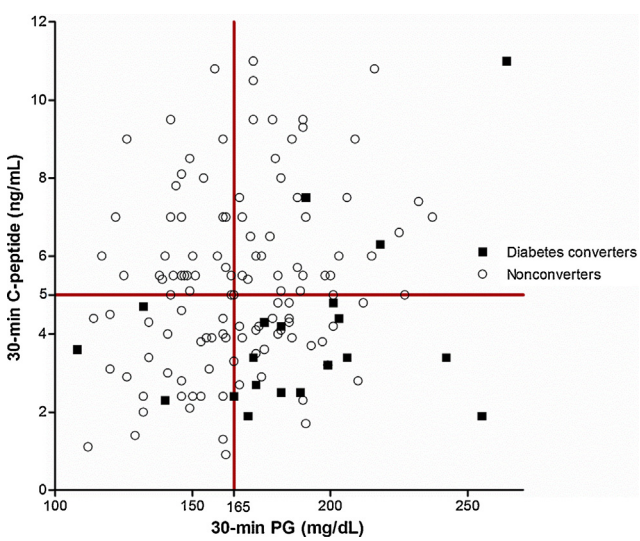

Fig. 2 – Scatter plot showing the incidence of type 2 diabetes according to 30-min plasma glucose and 30-min C-peptide. The subjects who had all the values of plasma C-peptide, insulin, and glucose at fasting, 30 min, and 2 h were included (*n* = 141). White circles = nonconverters; black squares = converters. Reference line indicates 30-min plasma glucose (PG) = 165 mg/dL, 30-min C-peptide = 5 ng/mL; Subjects with 30-min PG ≥165 mg/dL and 30-min C-peptide <5 ng/mL are a high-risk group for type 2 diabetes (HR = 8.83, 95% CI 2.98–26.16, *P* < 0.01), after adjustment for age, sex, BMI, family history of type 2 diabetes, and smoking. There was no converter in the group with 30-min PG <165 mg/dL and 30-min C-peptide ≥5 ng/mL.

In this study, the subjects with CGI and isolated IGT had a higher risk of diabetes than those with isolated IFG or isolated elevated HbA1c without IFG or IGT. Several epidemiologic studies reported that CGI and IGT were associated with a higher diabetes conversion rate than isolated IFG [33–37]. It has been shown that IGT is more closely associated with β-cell dysfunction than is IFG [29,30,38]. Both IFG and IGT have impaired early-phase insulin secretion, while IGT is associated with a greater reduction in late-phase insulin release [2,10,23,34]. Interestingly, the subjects with HbA1c 5.7–6.4% (39–46 mmol/mol) showed impaired early-phase insulin secretion but not impaired late-phase secretion [39] and lower diabetes risk than those with IGT [37]. These results suggest that a more pronounced β-cell dysfunction in subjects with IGT than in those with IFG and elevated HbA1c ≥5.7% (39 mmol/mol) might be associated with a higher risk of diabetes progression.

The results of an OGTT reflect a more physiologic impaired glucose homeostasis; hence, it would be an appropriate evaluation tool for future diabetes among subjects with prediabetes. During the OGTT, glucose and C-peptide levels at 30 min rather than fasting or 2 h had the highest predictive value for the progression to diabetes and showed a synergistic effect [40]. After glucose loading, first-phase insulin release occurs within 30 min and plays an important role in the suppression of glucose production in the liver [23,40,41]. Impaired first-phase insulin secretion leads to further hyperglycemia, which alters the glucose sensitivity of β-cells [42]. Late-phase insulin release and peripheral insulin sensitivity would affect 2-h PG. In our and previous studies, 2-h PG showed a weaker predictive power than 30-min PG or 1-h PG [15,34]. Individual plasma glucose and insulin levels assessed before the hyperbolic interaction between them might be a more accurate, independent predictor of incident diabetes in people with prediabetes.

Recent studies have shown that 1-h PG during the OGTT is a better predictor for diabetes than fasting glucose or 2-h PG [15,34,43]. Both 30-min PG and 1-h PG showed good correlations with indices of β-cell function, while FPG or HbA1c did

Table 5 – β-Cell function and insulin sensitivity-related indices among subjects with prediabetes divided into four groups by 30-min PG of 165 mg/dL and 30-min C-peptide of 5 ng/mL (*n* = 141).

	30-min PG <165 mg/dL and 30-min C-peptide ≥5 ng/mL (*n* = 30)	30-min PG <165 mg/dL and 30-min C-peptide <5 ng/mL (*n* = 31)	30-min PG ≥165 mg/dL and 30-min C-peptide ≥5 ng/mL (*n* = 41)	30-min PG ≥165 mg/dL and 30-min C-peptide <5 ng/mL (*n* = 39)	*P* value
Stumvoll index					
First phase insulin release	1375.3 ± 456.6	855.0 ± 331.8	1090.4 ± 548.1	557.0 ± 315.2	<0.01[c]
Second phase insulin release	359.5 ± 106.0	237.7 ± 77.1[b]	308.9 ± 125.8	179.8 ± 74.3	<0.01[c]
HOMA-β	231.0 ± 218.7	140.8 ± 88.4	149.9 ± 78.7	102.5 ± 55.7	<0.01[c]
Insulinogenic index (30′)	0.99 ± 0.48	0.44 ± 0.33	0.62 ± 0.33	0.28 ± 0.18	<0.01[c]
Insulinogenic index (120′)[a]	0.51 ± 0.26	0.25 ± 0.15	0.38 ± 0.17	0.22 ± 0.11	<0.01[c]
Disposition index	0.36 ± 0.26	0.19 ± 0.15	0.18 ± 0.09	0.10 ± 0.06	<0.01[c]
Glucose to insulin ratio	7.8 ± 4.4[*]	10.3 ± 4.6	7.8 ± 3.0[†]	11.8 ± 7.2	<0.01[b]
HOMA-IR	3.4 ± 1.6	2.5 ± 1.4	3.9 ± 2.0[*,‡]	2.8 ± 1.2	<0.01[b]
QUICKI	0.33 ± 0.02	0.34 ± 0.02	0.32 ± 0.18[*,‡]	0.33 ± 0.03	<0.01[b]
Matsuda index[a]	3.4 ± 1.7	6.1 ± 3.2	3.0 ± 1.2	4.9 ± 3.2	<0.01[c]

[*] $P < 0.05$ vs. subjects with 30-min PG ≥165 mg/dL and 30-min C-peptide <5 ng/mL.
[†] $P < 0.01$ vs. subjects with 30-min PG ≥165 mg/dL and 30-min C-peptide <5 ng/mL.
[‡] $P < 0.01$ vs. subjects with 30-min PG <165 mg/dL and 30-min C-peptide <5 ng/mL.
[a] Calculated from values at 0,30, and 120 min of the OGTT, using trapezoidal rule.
[b] Using ANOVA.
[c] Using Kruscal–Wallis tests, because they were not normally distributed.

not [34]. Hayashi et al. have reported that the patterns of insulin levels during an OGTT can predict the development of diabetes [16]. They measured 30-, 60-, and 120-min insulin levels during the OGTT and found that patients with an early insulin peak had a lower risk for diabetes than those with a later insulin peak. After 30 min, the values for insulin concentrations became diverse and were hard to interpret and generalize. Similarly, in our study, only 30-min C-peptide and 30-min insulin differed significantly between diabetes converters and nonconverters, while insulin and C-peptide at fasting and 2 h did not.

This, to the best of our knowledge, is the first demonstration that a combination of glycemic and insulin parameters from an OGTT could be a good predictor for diabetes progression. The predictive model with 30-min PG and 30-min C-peptide is convenient to perform in the clinical setting, with a shorter duration than the 2-h OGTT, and is easy to interpret without further calculation. In addition, these values can provide physicians with information about the underlying pathophysiology, either impaired β-cell function or increased insulin resistance, in subjects with prediabetes. With these categories, patients can be guided to receive tailored treatment. We may want to use the finding that individuals with CGI and isolated IGT have a higher risk of incident diabetes than those with isolated IFG or isolated elevated HbA1c in consultations about their future diabetes risk.

This study has some advantages. The long-term study period is the strength of this study. The follow-up rate was high, which ensured good validity for the results. Dynamic measures of glucose regulation also enhanced the prognostic value for progression to diabetes.

This study also has some limitations. First, the study population was entirely Korean; people of different ethnicity may have a different composition of types of prediabetes, and this difference might play a different role in progression to diabetes. Second, the glycemic and insulin-related parameters were measured only at fasting, 30 min, and 2 h.

In conclusion, a higher risk for future diabetes in patients with isolated IGT and CGI than in those with isolated IFG or isolated elevated HbA1c could be related to the idea that β-cell dysfunction is the major determinant for diabetes conversion in Asians, or at least in Koreans. The combination of 30-min PG and 30-min C-peptide values may be able to be used to assess the potential for future diabetes, considering its high predictive power for conversion to diabetes. Given that this finding has been validated in other studies, different levels of intervention can be applied to different categories of prediabetes for prevention of future diabetes.

Conflict of interest

The authors have no potential conflict of interest to declare.

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Please cite this article in press as: Kim YA, et al. Role of various indices derived from an oral glucose tolerance test in the prediction of conversion from prediabetes to type 2 diabetes. Diabetes Res Clin Pract (2014), http://dx.doi.org/10.1016/j.diabres.2014.08.014

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Please cite this article in press as: Kim YA, et al. Role of various indices derived from an oral glucose tolerance test in the prediction of conversion from prediabetes to type 2 diabetes. Diabetes Res Clin Pract (2014), http://dx.doi.org/10.1016/j.diabres.2014.08.014